



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA1522

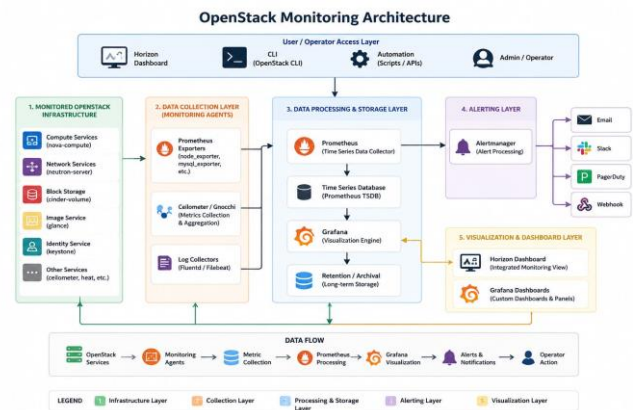
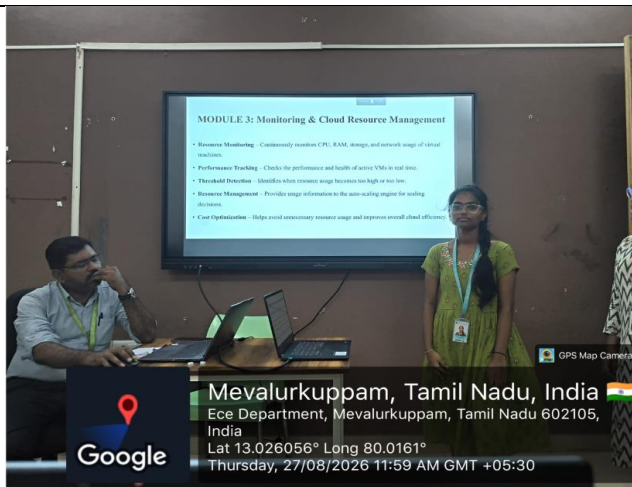
Slot: C

Course Name: Cloud Computing and Big Data Analytics

Course Faculty: Dr. P. Rajaram

Project Title: Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System

Module Photographs:



Project Description:

In this project, I developed a **cloud-based IaaS virtual infrastructure provisioning and auto-scaling system** to automatically manage computing resources according to workload requirements. I designed the model to provision virtual machines and allocate resources based on user demand. The system monitors parameters such as CPU utilization, memory usage, and workload, and automatically increases resources when demand is high and reduces them when demand decreases. I worked on defining scaling thresholds, provisioning logic, and scale-out and scale-in operations. I also tested the system under different workload conditions to evaluate its performance and resource utilization. The model helps improve scalability, availability, and cost efficiency by providing resources only when required. Overall, my contribution focused on developing an automated and flexible cloud infrastructure management approach using IaaS and auto-scaling concepts.

Student Signature

Guide Signature